

MOTION IN A STRAIGHT LINE

UNIT 9 - COMPLETE BATCH

Study Guides followed by ExamSIDE Transfer Books and solutions.

M-ST09A Relative acceleration: common gravity cancels

2 primary

M-ST09B Dropped from a moving platform: inherit carrier velocity

1 primary

M-ST09C Upward vs downward launch: same impact speed

1 primary

M-ST09D Tower launch time gap and geometric-mean drop time

2 primary

BATCH AUDIT

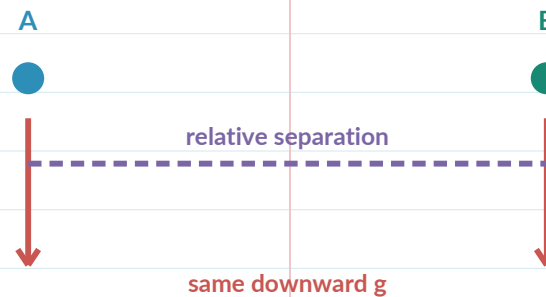
Current-primary ExamSIDE allocation: 6 required / 6 placed / 0 missing / 0 duplicate-primary.

Source anchors: Q39, Q51, Q52, Q62, Q46, Q54, Q47.

MOTION IN A STRAIGHT LINE

M-ST09A - Relative Acceleration: Common Gravity Cancels

Gravity still acts on both bodies; it disappears only from the separation equation.



MEMORY ANCHOR

SAME g ON BOTH $\rightarrow$ RELATIVE $a = 0$. Their relative velocity stays constant while both are in free flight.

SOURCE FOCUS

Source anchors: Q39, Q51 and related two-body items.

WHY GRAVITY CAN DISAPPEAR FROM A RELATIVE EQUATION

Subtract equal accelerations

M9A-C1

Define relative position

$$\mathbf{x}_{A/B} = \mathbf{x}_A - \mathbf{x}_B$$

$$\mathbf{a}_{A/B} = \mathbf{a}_A - \mathbf{a}_B$$

$$(-g) - (-g) = 0$$

IMPORTANT

Gravity did not vanish physically. It affects both bodies equally, so it cancels only when we describe their separation.

What follows

CONSTANT RELATIVE VELOCITY

Once both are freely falling under the same g , their relative velocity no longer changes.

FIRST MOVE

If two bodies are in the same gravitational field, try a relative-motion view before writing two long quadratic equations.

DELAYED LAUNCH: COMMON CLOCK + RELATIVE MOTION

Two ideas combine

M9A-C2

At the second launch

Ball 1 active



Ball 2 starts

STEP 1

Find the first ball's position and velocity at the instant the second ball is launched.

After both are in flight

$$a_{\text{rel}} = 0$$

STEP 2

Use constant relative velocity to close the existing separation. This avoids solving two full position quadratics from scratch.

CONNECTS BACK

M-ST06A supplies the delayed-start clock; M-ST09A supplies the relative-motion simplification.

MISCONCEPTION CLINIC: $a_{\text{rel}}=0$ DOES NOT MEAN BOTH BODIES HAVE $a=0$

Relative is not absolute

M9A-C3

Wrong idea

WRONG IDEA

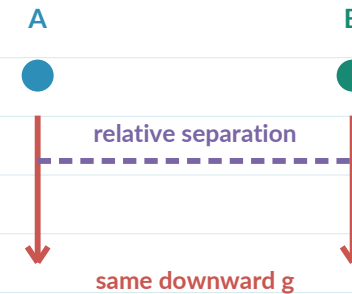
"Relative acceleration is zero, so gravity is not acting."

Each body still accelerates downward at g . Only the difference of their accelerations is zero.

CHECK

In the ground frame both velocity values keep changing. Their difference stays constant.

Correct model



LANGUAGE

Absolute acceleration: $-g$ for each. Relative acceleration: 0 between them.

M9A-G

Guided

Ball 1 is projected upward at 50 m/s. Two seconds later Ball 2 is projected upward from the same point at 50 m/s. Take $g=10$. Find when Ball 2 meets Ball 1.

H1 -> H2 -> H3

H1: at $t=2$, find Ball 1's position and speed. H2: after $t=2$, $a_{\text{rel}}=0$. H3: relative speed is constant; close the separation.

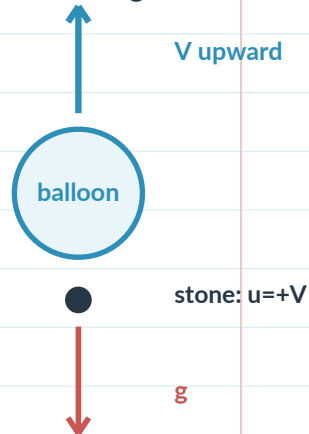
Self-check

- 1 form $x_{A/B}$ and $a_{A/B}$
- 2 recognise equal g cancellation
- 3 distinguish absolute and relative acceleration
- 4 combine delayed start with relative motion
- 5 use constant relative speed only after both share the same acceleration

MOTION IN A STRAIGHT LINE

M-ST09B - Dropped from a Moving Platform

“Dropped” means no extra throw relative to the carrier - not zero ground velocity.



MEMORY ANCHOR

RELEASED FROM MOVING CARRIER -> INHERIT CARRIER VELOCITY.

SOURCE FOCUS

Source anchors: Q52 and Q62.

WHAT “DROPPED” REALLY MEANS

Reference frame first

M9B-C1

Relative to the balloon

$v_{\text{stone/balloon}} = 0$ at release

NO EXTRA THROW

The hand does not give the stone a new velocity relative to the balloon.

Relative to the ground

$u_{\text{stone}} = V_{\text{balloon}}$

IF BALLOON RISES

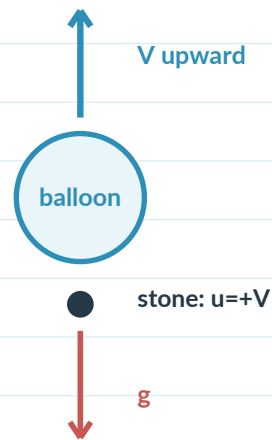
With upward positive and balloon speed $+V$, the stone begins free flight with $u=+V$ and $a=-g$.

THE STONE FIRST KEEPS GOING UP

Inertia before reversal

M9B-C2

Immediately after release



Equations in ground frame

$$y = y_0 + Vt - \frac{1}{2}gt^2$$

$$v = V - gt$$

DO NOT RESET $u=0$

The free-flight stage begins with the carrier's instantaneous velocity.

STORY

The stone is already moving upward. Gravity slows that upward motion, brings v to zero, then makes it fall.

M9B-C3

Wrong idea

WRONG IDEA

"Dropped means $u=0$."

That is true only when the releasing hand/platform is at rest in the chosen frame.

COUNTEREXAMPLE

A stone released from a rising balloon has upward ground velocity at the instant of release.

Correct helper

FRAME $\rightarrow$ CARRIER $\rightarrow$ OBJECT

1. Choose the observer/frame. 2. Read carrier velocity in that frame. 3. Add any release velocity relative to the carrier. For a simple drop, the relative term is zero.

M9B-G

Guided

A balloon rises at 10 m/s and releases a stone at height 75 m. Take $g=10$. Write the stone's ground-frame initial velocity and the equation for its height after t seconds.

H1 -> H2 -> H3

H1: ground frame. H2: $u=+10$. H3: $y=75+10t-5t^2$.

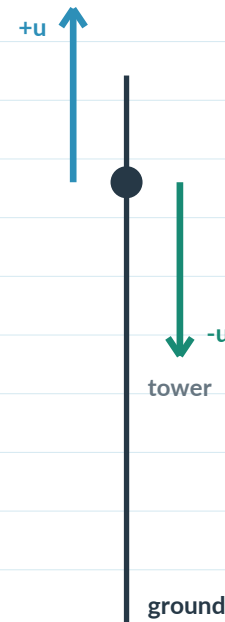
Self-check

- 1 distinguish relative and ground velocity
- 2 inherit carrier velocity at release
- 3 explain why the stone can initially keep rising
- 4 write $y(t)$ with the inherited u
- 5 avoid assuming every 'drop' begins from rest

MOTION IN A STRAIGHT LINE

M-ST09C - Upward vs Downward Launch: Same Impact Speed

Initial direction changes the journey, but not the final speed magnitude at the same lower level.



MEMORY ANCHOR

SAME $|u|$ + SAME HEIGHT DROP $\rightarrow$ SAME final speed magnitude.

SOURCE FOCUS

Source anchors: Q46 and Q54.

WHY THE FINAL SPEEDS MATCH

The no-time equation removes the sign of u

M9C-C1

Both start at the same tower top

$$v_f^2 = u^2 + 2gH$$

KEY DETAIL

Whether the initial velocity is $+u$ or $-u$, its square is u^2 .

What does change?

PATH HISTORY

The upward-thrown ball rises first, returns, then falls. The downward-thrown ball begins falling immediately.

WHAT DOES NOT CHANGE

At the common ground level, ideal final speed magnitudes are equal.

MISCONCEPTION CLINIC: LONGER PATH DOES NOT MEAN GREATER FINAL SPEED

Height change controls v^2

M9C-C2

Wrong idea

WRONG IDEA

"The upward-thrown ball travels farther, so it must hit faster."

In this ideal constant-g model, the speed-at-height relation depends on initial speed magnitude and net vertical displacement, not total path length.

WHY IT FEELS PLAUSIBLE

Distance travelled often matters in resistive or energy-loss situations. Here, air resistance is neglected.

Correct model

$$v_f = \sqrt{u^2 + 2gH}$$

MODEL GATE

If air resistance is important, the path can matter and this equality need not hold.

M9C-G

Guided

From the same tower top, one ball is thrown upward at 12 m/s and another downward at 12 m/s. Ignore air resistance. Compare their speeds at the ground.

H1 -> H2 -> H3

H1: same $|u|$. H2: same net height H . H3: write $v_f^2 = u^2 + 2gH$ for each.

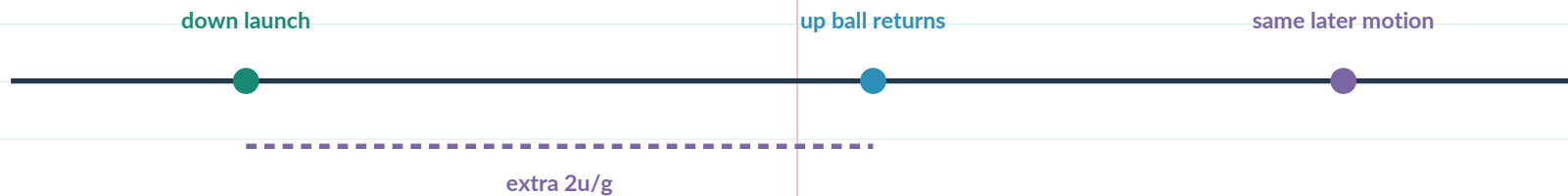
Self-check

- 1 separate speed magnitude from velocity sign
- 2 use the no-time relation for same endpoint height
- 3 explain why u^2 removes launch direction
- 4 distinguish path length from net displacement
- 5 state the no-air-resistance assumption

MOTION IN A STRAIGHT LINE

M-ST09D - Time Gap: Upward vs Downward Launch from a Tower

The upward launch contains one extra symmetric excursion.



MEMORY ANCHOR

EXTRA TIME = $2u/g$. The upward ball rises and returns to launch level before matching the downward ball's starting state.

SOURCE FOCUS

Source anchor: Q47.

WHY THE TIME DIFFERENCE IS SIMPLE

Shift the same downward journey

M9D-C1

Downward-launched ball

STARTS THE DOWNWARD JOURNEY NOW

Initial downward speed magnitude u from the tower top.

Once it returns

SAME STATE

At the tower top on the way down, it has speed magnitude u downward - exactly the state of the other ball at its launch.

Upward-launched ball

EXTRA LOOP

It first spends u/g rising and u/g returning to the tower top.

$$\Delta T = 2u/g$$

M9D-C2

Let times be t_1 and t_2

From the same tower, equal speed magnitude u is used once upward and once downward. Let their ground-hit times be t_1 and t_2 .

$$t_1 - t_2 = 2u/g$$

SAME HEIGHT EQUATIONS

Combining the two tower-height equations also gives the drop-from-rest time t_0 .

Result

$$t_0 = \sqrt{t_1 t_2}$$

WHY IT IS USEFUL

You can find the release-from-rest time without first solving separately for tower height and launch speed.

MISCONCEPTION CLINIC: UPWARD TIME IS NOT JUST DOWNWARD TIME + u/g there are two extra legs

M9D-C3

Wrong idea

WRONG IDEA

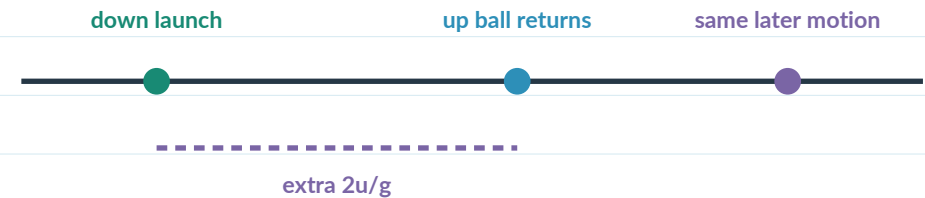
"The upward launch only adds the time to reach the top, u/g ."

It must also spend another u/g returning from the top to the original launch level.

TWO SYMMETRIC LEGS

rise to top + return to launch level = $u/g + u/g$.

Correct model



STATE MATCHING

After $2u/g$, the upward-launched ball has the same position and downward speed as the downward-launched ball had initially.

M9D-G

Guided

A ball thrown upward from a tower takes 6 s to reach the ground. Thrown downward from the same point with the same speed, it takes 1.5 s. Find the time if simply dropped.

H1 -> H2 -> H3

H1: use the paired-time relation. H2: $t_0 = \sqrt{t_{\text{up}} t_{\text{down}}}$. H3: $\sqrt{6 \cdot 1.5}$.

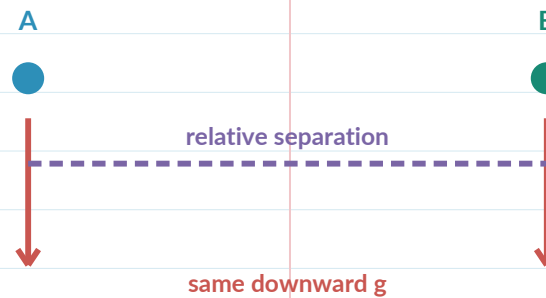
Self-check

- 1 explain the extra symmetric $2u/g$ excursion
- 2 recognise the same-state time shift
- 3 use $t_1 - t_2 = 2u/g$
- 4 use $t_0 = \sqrt{t_1 t_2}$ when the setup matches
- 5 avoid using the result for unequal speed magnitudes

MOTION IN A STRAIGHT LINE

M-ST09A - ExamSIDE Transfer Book

Two free-flight bodies under the same g .



CURRENT PRIMARY SET

Required = 2 • placed = 2 • missing = 0.

DIFFICULTY

D3-D4: both require delayed-start translation plus relative-motion simplification.

Q1

D3

M9A

Question

A ball is projected upward at 50 m/s at $t=0$. Another is projected upward at 50 m/s at $t=2$ s. Find the meeting time, $g=10$.

Concept link

STUDY GUIDE

M-ST09A - equal gravitational acceleration cancels in the relative equation.

H1 / H2 / H3

At $t=2$, Ball 1 is 80 m above launch and moving at 30 m/s. Ball 2 starts at 50 m/s. After that, $a_{\text{rel}}=0$ and closing speed is 20 m/s.

[Open ExamSIDE source](#)

Q2

D4

M9A

Question

Two equal spherical balls are thrown upward 3 s apart with the same initial velocity 35 m/s. Each radius is 5 cm. Find the collision height, $g=10$.

Concept link

STUDY GUIDE

M-ST09A - equal gravitational acceleration cancels in the relative equation.

H1 / H2 / H3

At the second launch, compute the first ball's state. Then use constant relative speed, while remembering collision occurs when centres are 10 cm apart; the published integer result is 50 m.

[Open ExamSIDE source](#)

A

EX-M9A-01

SOLUTION

At $t=2$ s, Ball 1: $y=50(2)-5(4)=80$ m and $v=50-20=30$ m/s upward. Ball 2 begins at 50 m/s, so relative closing speed is 20 m/s and $a_{\text{rel}}=0$. Time after second launch $=80/20=4$ s. Global meeting time = 6 s.

EX-M9A-02

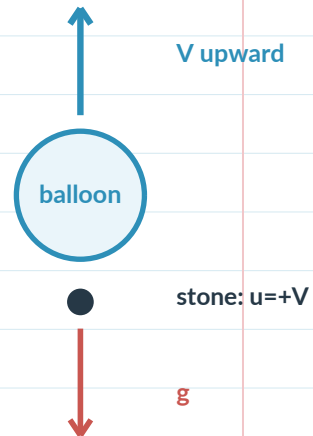
SOLUTION

At the second launch (3 s), Ball 1 is at 60 m and moving upward at 5 m/s. Ball 2 starts upward at 35 m/s, so centre-to-centre closing speed is 30 m/s and remains constant. Allowing for the 10 cm centre separation at contact gives a collision just before 2 s after the second launch; the corresponding height rounds to the published integer 50 m.

MOTION IN A STRAIGHT LINE

M-ST09B - ExamSIDE Transfer Book

Inherited velocity at release from a moving carrier.



CURRENT PRIMARY SET

Required = 1 · placed = 1 · missing = 0.

DIFFICULTY

D2: the decisive step is the reference-frame initial condition.

Q1

D2

M9B-C1

Question

A gas balloon rises at 10 m/s. At height 75 m a stone is dropped, while the balloon continues upward at the same speed. Find the balloon's height when the stone hits the ground, taking $g=10$.

Concept link

STUDY GUIDE

M-ST09B - dropped relative to the balloon means inherited ground velocity, not $u=0$.

H1 / H2

Stone starts with $u=+10$ m/s in the ground frame. Solve $0=75+10t-5t^2$; then advance the balloon by $10t$.

[Open ExamSIDE source](#)

A

Solution

FULL ROUTE

Use upward positive. Stone: $y=75+10t-5t^2$. Setting $y=0$ gives $t=5$ s. During those 5 s, the balloon rises another 50 m. Its height is $75+50=125$ m.

Counters

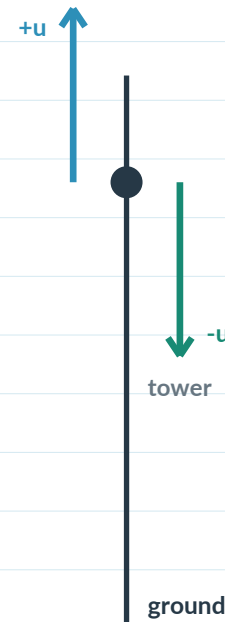
AUDIT

PRIMARY REQUIRED 1 • PLACED 1 • MISSING 0 • SOLUTIONS 1/1.

MOTION IN A STRAIGHT LINE

M-ST09C - ExamSIDE Transfer Book

Equal initial speed magnitude, same tower, same ground level.



CURRENT PRIMARY SET

Required = 1 • placed = 1 • missing = 0.

SOURCE RANGE

ExamSIDE chapter history includes AIEEE 2002 within the JEE Main 2002-2026 archive.

EX-M9C-01 - TWO BALLS FROM A BUILDING

AIEEE 2002 - adapted

Q1

D2

M9C-C1

Question

From a building, Ball A is thrown upward and Ball B downward with the same speed magnitude. Compare their velocities on reaching the ground, neglecting air resistance.

Concept link

STUDY GUIDE

M-ST09C - initial direction changes time/path, not the ideal final speed magnitude.

H1 / H2

Their velocity directions at impact are both downward. For speed magnitude, use $v^2 = u^2 + 2gH$ for each.

[Open ExamSIDE source](#)

A

Solution

FULL ROUTE

Both balls begin with the same speed magnitude u at the same height H above ground. For either ball, the final speed satisfies $v^2 = u^2 + 2gH$. Therefore the impact speed magnitudes are equal. Since both are moving downward at impact, their velocity vectors are also equal in this one-dimensional setup: $v_A = v_B$.

Counters

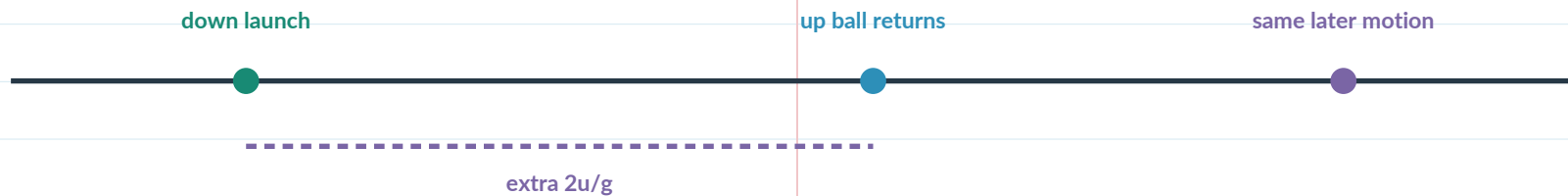
AUDIT

PRIMARY REQUIRED 1 • PLACED 1 • MISSING 0 • SOLUTIONS 1/1.

MOTION IN A STRAIGHT LINE

M-ST09D - ExamSIDE Transfer Book

Upward/downward tower launch time relations.



CURRENT PRIMARY SET

Required = 2 • placed = 2 • missing = 0.

DIFFICULTY

D2-D3: one symbolic paired-time question and one numerical version.

Q1

D3

M9D

Question

A body thrown upward from a tower reaches ground in t_1 . Thrown downward with the same speed it reaches in t_2 . Find the time if simply dropped.

Concept link

STUDY GUIDE

M-ST09D - upward launch is the downward-launch journey shifted later by the symmetric $2u/g$ excursion.

H1 / H2

Use the paired tower equations or the derived shortcut $t_0 = \sqrt{t_1 t_2}$.

[Open ExamSIDE source](#)

Q2

D2

M9D

Question

From a tower, upward launch reaches ground in 6 s and equal-speed downward launch in 1.5 s. Find the drop-from-rest time.

Concept link

STUDY GUIDE

M-ST09D - upward launch is the downward-launch journey shifted later by the symmetric $2u/g$ excursion.

H1 / H2

Apply $t_0 = \sqrt{6 \times 1.5}$.

[Open ExamSIDE source](#)

A

EX-M9D-01

SOLUTION

Using the same tower height for upward and downward launches leads to the standard paired-time result. For a simple drop, $t_0 = \sqrt{t_1 t_2}$.

EX-M9D-02

SOLUTION

$$t_0 = \sqrt{(6 \times 1.5)} = \sqrt{9} = 3 \text{ s.}$$